

Single-head Gummiworm protocol:

Timing rules

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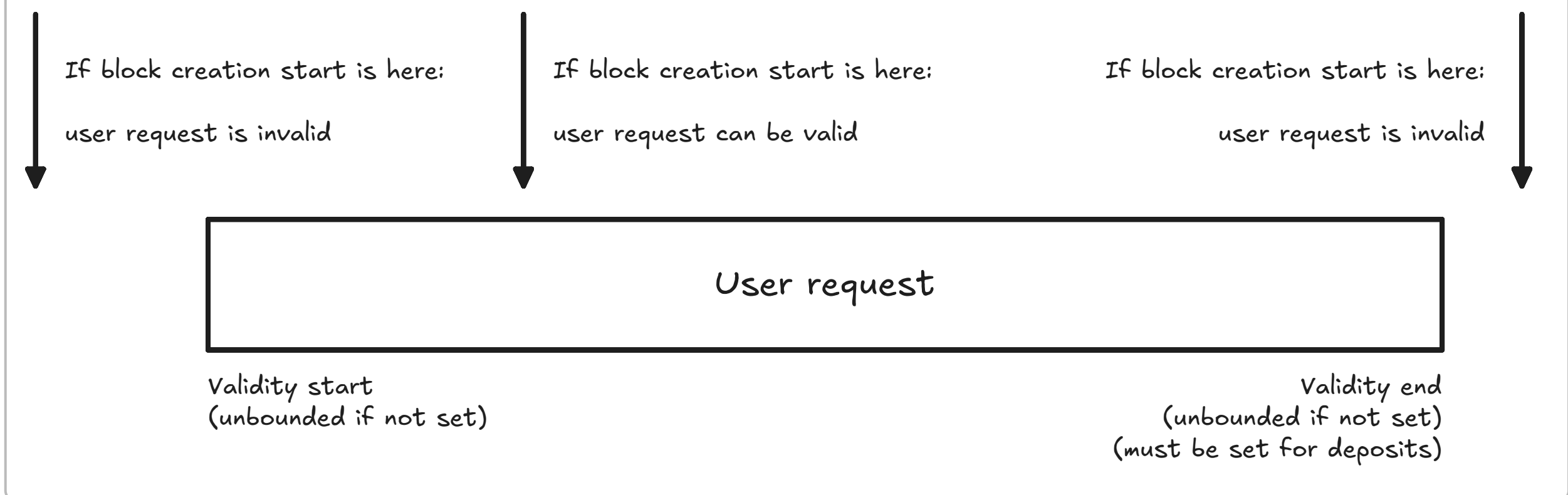
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User requests

A block brief can declare a user request to be valid only if the block's creation start time falls within the request's validity interval.

User request validity bounds are optional for transaction requests, but the validity upper bound is mandatory for deposit requests.

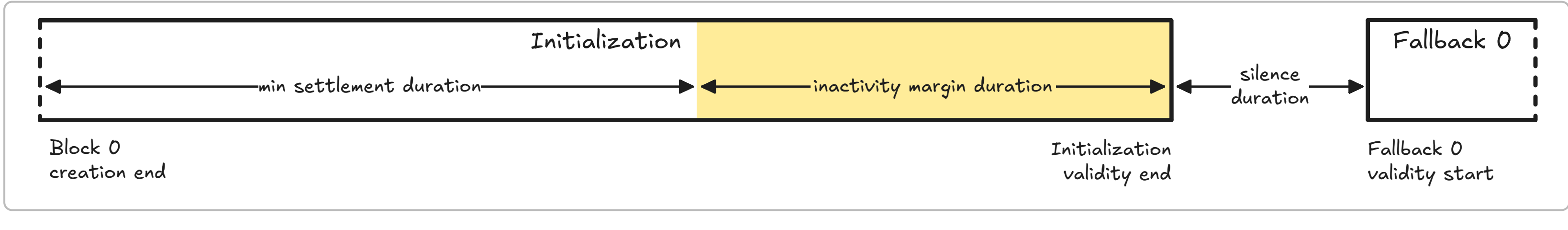
When a lower or upper validity bound is not set for a user request, its validity interval is unbounded on that side.



Initialization and initial fallback

Given the block 0 creation time:

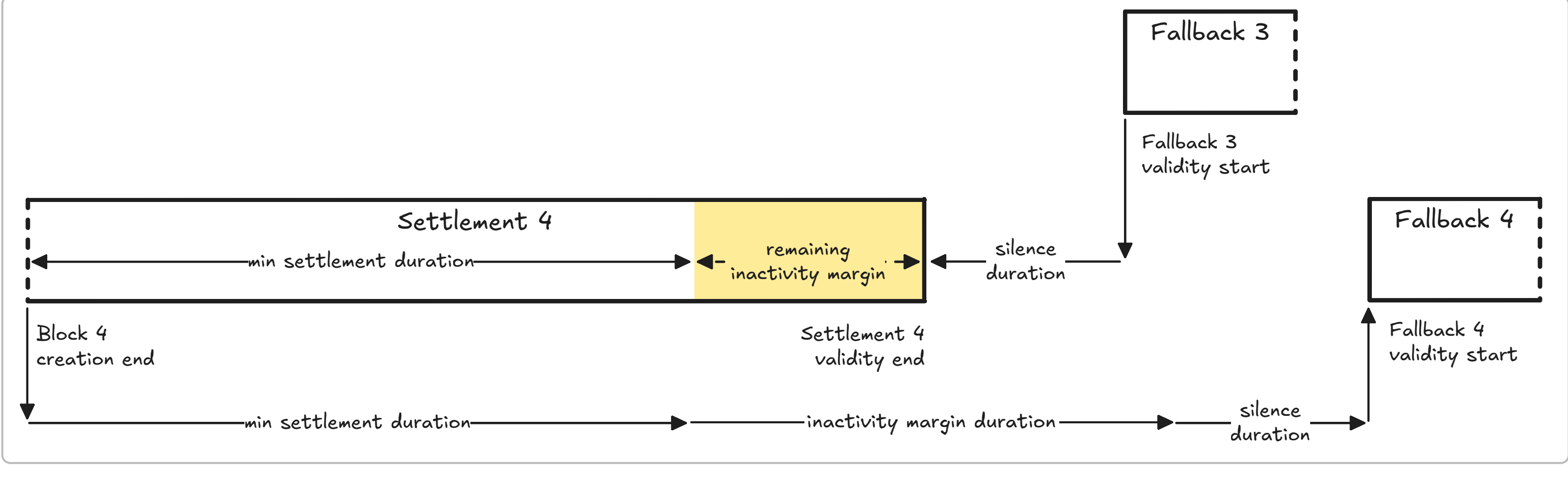
- The initialization effect's validity end time is offset by the min settlement and inactivity margin durations.
- The initial fallback effect's validity start time is further offset by the silence duration.



Settlement and fallback

Each settlement effect's validity end is offset backward from the previous fallback's validity start time by the silence duration.

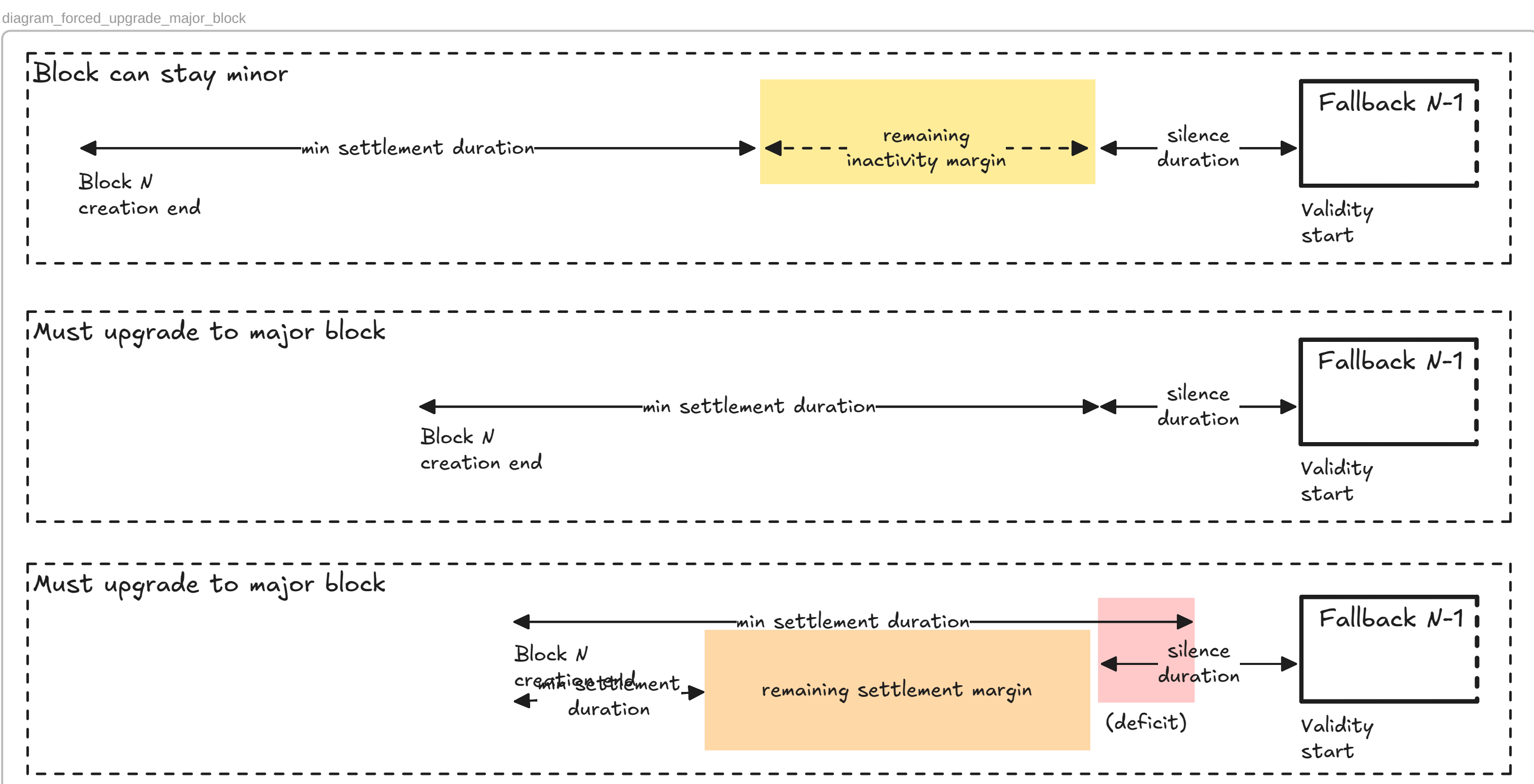
Each fallback effect's validity start time is offset (forward) from its corresponding block's creation time by the min settlement, inactivity margin, and silence durations.



Forced upgrade to major block

A minor block must be upgraded to a major block if too little time remains before the competing fallback validity start.

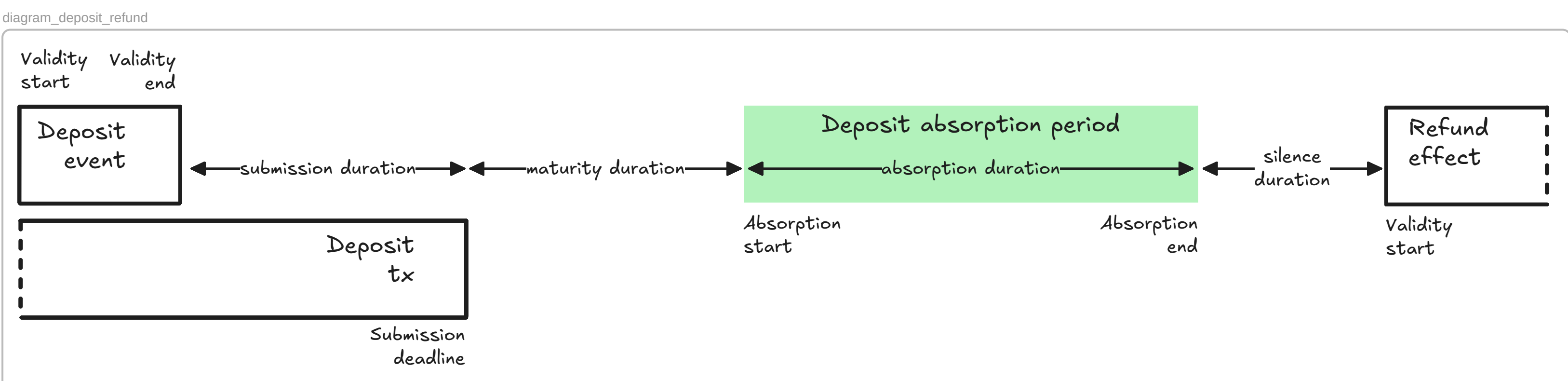
This causes new settlement and fallback effects, which supersede the old fallback with a later fallback validity start.



Deposit and refund

Relative to a deposit request's validity end:

- the deposit's submission deadline is offset by the submission duration.
- the deposit's absorption start time is further offset by the deposit maturity duration.
- the deposit's absorption end time is further offset by the absorption duration.
- the refund effect's validity start time is further offset by the silence duration.

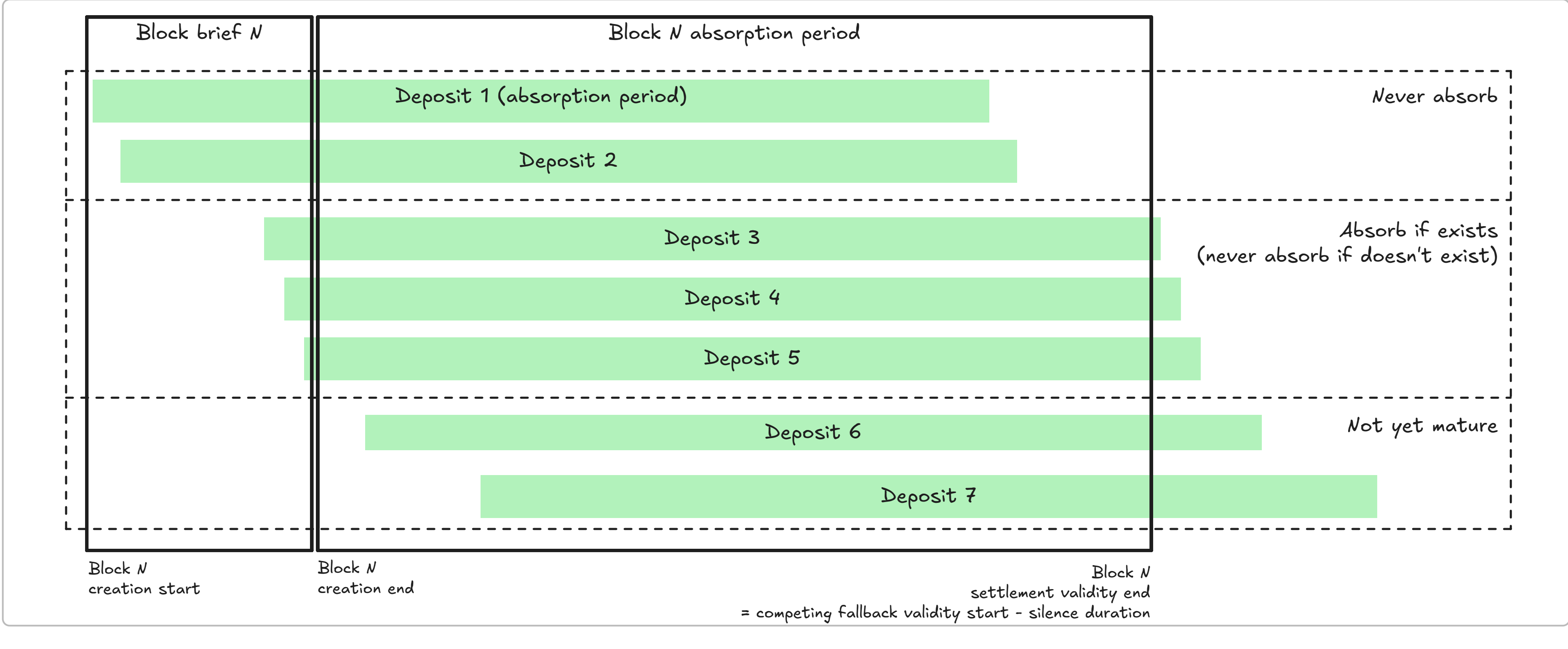


Deposit absorption decision

The decision of whether to absorb or reject a deposit depends on these two factors:

- Whether the block's absorption period falls within the deposit's absorption period.
- Whether the fast-consensus can see the deposit (after it matures) on the L1 blockchain at the block's creation end time.

A block's absorption period starts at its creation end time and ends at its settlement validity end time (notional for minor blocks).



Suggested parameter values

